

Week 4 Report

PENTAS HUB Internship Program

Project Title

Diabetes & Heart Disease Risk Predictor Using Transformer-Based Medical Text Processing

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Week Number: Week 4

Domain: Machine Learning / Natural Language Processing (NLP)

Duration: Week 4

1. Introduction

During Week 4 of the internship, the project was extended by integrating transformer-based Natural Language Processing (NLP) techniques for processing patient notes and medical text. A pre-trained BERT model was used to understand and analyze textual medical information, enabling the system to interpret patient symptoms more effectively alongside structured clinical data.

2. Objectives

- Process patient notes using transformer models.
 - Perform medical text preprocessing.
 - Generate contextual embeddings using BERT.
 - Analyze symptoms from medical text.
 - Integrate text processing into the disease prediction workflow.
 - Prepare the project for explainable AI implementation.
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3. Tasks Completed

Medical Text Collection

- Collected sample patient notes.
- Prepared medical symptom descriptions.
- Organized text data for processing.

Text Preprocessing

- Removed unnecessary symbols.
- Converted text into a suitable format.
- Tokenized medical sentences.
- Prepared input for the transformer model.

Transformer Model Implementation

- Implemented a pre-trained BERT model.
 - Generated contextual embeddings.
 - Processed patient notes for disease-related information.
 - Integrated text analysis into the prediction pipeline.
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4. Technologies Used

- Python
 - Hugging Face Transformers
 - BERT
 - PyTorch
 - Pandas
 - Google Colab
 - GitHub
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5. Files Developed

Source Files

- text_preprocessing.py
- bert_medical_text.py
- medical_text_prediction.py
- evaluate_transformer.py

Datasets

- patient_notes.csv
- medical_text_dataset.csv

Model Files

- bert_medical_model.pkl
 - tokenizer.pkl
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6. Evaluation Metrics

The transformer model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score

The model successfully processed medical text and generated contextual representations that can support disease prediction.

7. Results

- Successfully implemented transformer-based medical text processing.
 - Processed patient notes using a BERT model.
 - Generated contextual embeddings for medical text.
 - Integrated NLP processing into the existing prediction workflow.
 - Prepared the system for an explainable AI chatbot in the next phase.
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8. Challenges Faced

- Understanding transformer architecture.
 - Preparing medical text for NLP processing.
 - Tokenizing patient notes correctly.
 - Integrating textual and structured medical data.
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9. Solutions Implemented

- Used a pre-trained BERT model from the Hugging Face Transformers library.
 - Applied tokenizer-based preprocessing.
 - Generated embeddings using transformer outputs.
 - Validated the processed text before prediction.
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10. Learning Outcomes

During Week 4, I learned:

- Fundamentals of Natural Language Processing (NLP).
 - Transformer architecture and BERT.
 - Medical text preprocessing techniques.
 - Tokenization and contextual embeddings.
 - Integrating transformer models into machine learning projects.
 - Processing patient notes for healthcare applications.
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11. Conclusion

Week 4 focused on incorporating transformer-based NLP into the disease prediction system. Medical text and patient notes were processed using a pre-trained BERT model, enabling the project to interpret textual health information alongside structured patient data. This enhancement improves the intelligence and flexibility of the system and prepares it for the development of an explainable AI chatbot in Week 5.

12. Future Work

- Develop a Retrieval-Augmented Generation (RAG) chatbot.
- Integrate the chatbot with the disease prediction models.
- Provide AI-generated explanations for prediction results.
- Deploy the complete application using Streamlit.
- Finalize the project documentation and presentation.